

CLEF CheckThat! Task 2 — Course Project Tracks

In this project, you will participate in CLEF CheckThat! Task 2 and work in teams on different methodological tracks.

All teams solve the same core problem:

Given a claim containing quantitative and/or temporal expressions, relevant evidence, and a set of LLM-generated reasoning traces with corresponding verdicts, train a verifier model that (1) ranks the reasoning traces by their utility in leading to the correct verdict, and (2) outputs a final verdict derived from the top-ranked traces.

Each team is assigned one track and must focus on that methodological direction.

General Requirements (All Teams)

0. Competition Registration (*Required*)

Failure to register means the project cannot be graded.

- Register for CLEF CheckThat! Task 2.
- Choose a team name.
- Include your team name in your report appendix.

1. Competitive Baseline (20%)

You must implement at least one strong baseline pipeline consisting of:

- A fine-tuned verifier model that ranks reasoning traces per claim
- A verdict aggregation strategy that derives a final label from the top-ranked traces
- Evaluation on development data and test data using macro F1 and Recall@5

This baseline should be competitive (consult proceedings from last year), reproducible, and well-documented.

2. Task-Specific Research & Experiments (50%)

Your main goal is to improve performance within your assigned track.

Expectations:

- Multiple meaningful experiments
- Clear motivation for design decisions
- Ablations and comparisons on training and test data
- Engagement with relevant literature
- Error analysis and discussion

The emphasis is on systematic experimentation and insight, not just running models.

3. Report (20%)

Submit a 2-page ACM-style (double column) report including:

- System description
- Experimental setup
- Results and discussion
- Work distribution

Appendix (*not part of page limit*):

- Team name
- Individual contributions: who worked on which parts of the system and experiments
- Additional result tables, runtimes, or ablations

4. Run Submission (10%)

Submit:

- TREC-formatted run files
- The same runs on the shared task platform

Runs must correspond to systems described in the report.

5. Code Submission (*Required*)

All teams must submit their code via GitHub Classroom. The repository must include:

- Instructions to reproduce runs
- Environment setup
- Scripts to generate submitted runs
- Model checkpoints or instructions to obtain them

The code must run with reasonable effort. Non-runnable code may result in loss of credit.

6. Exercise Interview (*Required — scales all grades*)

Each team member will participate in an individual oral interview with the course instructor after submission. The interview assesses personal understanding of the system, the design choices made, and the results obtained.

- The interview lasts approximately 15 minutes per person.
- Questions will cover your track's methodology, your team's experiments, and the broader task.
- Your individual interview performance acts as a **scaling factor** on your share of the team grade. A student who cannot explain their own system may receive a substantially lower individual grade even if the team submission is strong.
- Attendance is mandatory. Failure to attend without prior notice results in a grade of zero for the individual.

Track 1 — Small & Distilled Verifier Models (< 0.5B parameters)

Focus: Efficient and lightweight verifier models under strict size constraints.

Constraints:

- Verifier models must be below ~0.5 billion parameters.
- Model distillation from a larger teacher model is required.
- Avoid heavy cross-encoders or large models for the student verifier.

Goal:

Train a compact verifier model that ranks reasoning traces and predicts the final verdict accurately, while remaining deployable under real-world efficiency constraints.

You should evaluate:

- Inference latency and model size
- Performance vs. efficiency trade-offs (Recall@5, macro F1)
- The effect of different distillation strategies (response-based, feature-based, contrastive)

Track 2 — Verifier Training Objectives & Model Scale

Focus: Systematic comparison of learning-to-rank paradigms for reasoning trace verification, evaluated across model scales.

Allowed:

- Pointwise, pairwise, and listwise training objectives
- Models of varying sizes (use model scale as an experimental variable)
- Fine-tuning pretrained language models as verifiers

Goal:

Identify which training objective leads to the best reasoning trace ranker, and whether this finding holds consistently across model sizes.

You should evaluate:

- Pointwise vs. pairwise vs. listwise objectives in a controlled setting
 - Performance of each objective at multiple model scales (e.g., ~100M, ~500M, ~1B+ parameters)
 - Interaction effects: does the best objective change with scale?
 - Final verdict quality (macro F1) alongside ranking quality (Recall@5)
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Track 3 — Numerical & Temporal Reasoning Signals

Focus: Exploiting the quantitative and temporal structure of claims to improve trace ranking and verdict prediction.

Possible directions:

- Structured extraction of numbers, dates, and temporal expressions from claims and evidence
- Symbolic consistency checking between claim quantities and evidence quantities
- Numeric normalization and unit resolution as a pre-processing step
- Scoring traces based on numerical faithfulness to the evidence
- Augmenting the verifier with explicit numerical features or auxiliary tasks

Constraint:

The main contribution must come from reasoning signals derived from the quantitative or temporal content of claims — not from general model scaling.

Goal:

Improve trace ranking and verdict prediction by making the verifier sensitive to the specific numerical and temporal structure that defines this task.

Track 4 — Trace Aggregation & Verdict Derivation

Focus: How a final verdict is derived from a set of ranked reasoning traces.

Possible directions:

- Majority voting over top-k traces
- Confidence-weighted aggregation
- Meta-reasoning: using a model to synthesize the top-k traces into a single verdict
- Uncertainty estimation across traces
- Investigating how ranking quality (Recall@5) and verdict quality (macro F1) interact depending on aggregation strategy

Constraint:

Keep the verifier/ranker fixed or use the same baseline ranker across experiments. The main contribution must be in the aggregation step, not in the ranking model itself.

Goal:

Identify which aggregation strategy best converts a ranked list of reasoning traces into a correct and robust final verdict, and characterize when aggregation strategies fail.

Final Objective

Each team develops an independent system under their assigned track. At the end of the course:

- Systems will be compared across tracks on the shared test set
- Strengths and weaknesses will be analyzed jointly
- Systems may optionally be combined via ensemble methods

This project simulates a real shared-task research environment. The goal is not only to achieve high scores, but to produce reproducible, well-motivated, and clearly described research.